

1-Bit Full Adder Using Dataflow Modeling

Aim

To design and verify a 1-bit full adder in Verilog using dataflow modeling.

Code

```
module full_adder(  
    input A,  
    input B,  
    input Cin,  
    output Sum,  
    output Carry  
);  
  
assign Sum = A ^ B ^ Cin;  
assign Carry = (A & B) | (B & Cin) | (A & Cin);  
  
endmodule
```

Truth Table / Observation

A	B	Cin	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Observation: All 8 input combinations were simulated and matched the expected full-adder truth table. The simulation generated `full_adder.vcd` and completed at 80 ns.

Conclusion

The 1-bit full adder was successfully implemented using Verilog dataflow modeling. The `Sum` output correctly represents odd input parity, and the `Carry` output is high whenever at least two inputs are high.